

5.6 Proving Triangles Congruent

Lesson Introduction

 Benchmarks	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg. MA.912.LT.4.10 Judge the validity of arguments and given counterexamples to disprove statements.			 Learning Target	I can prove two triangles are congruent using SSS, SAS, ASA, AAS, or HL congruence.
 Benchmark Coherence	Building On N/A			Working Toward N/A	
 Essential Questions	- How can I prove that two triangles are congruent? - How can I determine which postulate or theorem best proves if two triangles are congruent?				
 Lesson Narrative	In this lesson, students extend their knowledge of triangle congruency to prove that two triangles are congruent. Students will complete two-column, flow, and paragraph style proofs in order to apply their knowledge from this unit.				
 Component Summary	5.6.1 Warm-Up (~5 min.)	Real-world estimation problem (<i>SE p. 253</i>)	5.6.3 Guided Practice (10 min.)	Practice with proving triangles congruent (<i>SE p. 255</i>)	
	5.6.2 Guided Instruction (15 min.)	Formal instruction on proving triangles congruent (<i>SE p. 253</i>)	5.6.4 Your Turn! (10 min.)	Independent practice proving two triangles are congruent (<i>SE p. 256</i>)	
	5.6.5 Wrap-Up (~5 min)	Error analysis of a flowchart proof with proving two triangles are congruent (<i>SE p. 256</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Colored Markers		- Students may struggle with the proofs in this lesson. Work through each one on your own prior to delivering instruction to work through potential misconceptions in the proofs. - Some proofs have alternate paths. Prepare knowing that students may consider these alternate paths.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not know the definitions used in the proofs. - Students may not realize all the information that is available in the image of a problem when it's not explicitly stated. - Student may apply the same marking to multiple sides, causing confusion.	Consider asking students to complete a chart, notebook, or foldable for the definitions, postulates, and theorems that will be used in proofs. This can include common reasons that usually follow each other. For example, "definition of perpendicular lines" is typically followed by "definition of right angles" or "right angles congruence theorem." Also, the definition of a bisector is typically followed by a congruence statement.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 5.6.3 Guided Practice has students completing a flow chart proof to prove two triangles are congruent. For this activity, students can think through their own steps and reasoning and then pair up with a partner to share their answers together to come to a consensus.	MA.K12.MTR.5.1: Patterns and Structure In 5.6.2 Guided Instruction, students will draw conclusions based on given information and explain why the conclusion can be drawn. Students will then order the steps to formally prove the conclusion from the given information.
 Differentiate Instruction	Consider placing students in groups based on their performance so far in this unit. Students will be applying their knowledge from prior lessons through the process of completing a proof. Students who struggled with triangle congruency will likely struggle proving it. For 5.6.2 Guided Instruction, consider using auditory explanations for a proof prior to filling it in. Use common language to describe how the triangles could be proven congruent. Then, transition that language into formal geometric language when completing the proof. The 5.6.5 Wrap-Up activity provides students with an opportunity to demonstrate their understanding of triangle congruence using a proof. The data provided from this activity can help determine which students may need additional support.	
Lesson Notes:		

5.6.1 Warm-Up: Cheese Balls

Component Overview: Students will engage in a real-world estimation problem. Students may need to be probed on what shapes they see in the figure and what measurements will be used.



5.6.1 Warm-Up: Cheese Balls

The image shows cheese balls arranged on a baking sheet with a white plate on top. There are thirteen cheese balls on the plate.



1. Estimate how many cheese balls will fit on the plate.

Answers will vary. 130 cheese balls.

2. Explain how you determined your estimate.

Answers will vary. If we estimate that the plate has a diameter of 13 cheese balls wide, we could estimate the area of the plate as $A = \pi(6.5^2) \approx 133$ cheese balls.



If you didn't want to use an estimation strategy and are asked for a specific answer, what information would you need to know in order to answer the question correctly? What would you do with the information to solve?

5.6.2 Guided Instruction: Proving Triangles Congruent

Component Overview: Students will receive guided instruction on proving triangles are congruent through a flow chart proof, two-column proof, and paragraph proof.

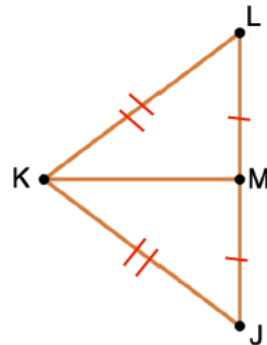


5.6.2 Guided Instruction: Proving Triangles Congruent

When completing a proof to show two triangles are congruent, it is helpful to start by considering conclusions that can be drawn from the given information.

1. Triangle KLJ is shown with some given information.

Given: $\overline{KM}$ bisects $\overline{LJ}$
 $\overline{KL} \cong \overline{KJ}$



- a. What conclusion can be drawn from the given statement, " $\overline{KM}$ bisects $\overline{LJ}$ "?

$$\overline{JM} \cong \overline{LM}$$

- b. Explain why this conclusion can be drawn.

Answers will vary. Because $\overline{KM}$ bisects $\overline{LJ}$, by the definition of segment bisector $\overline{LJ}$ is divided into two equal parts.

- c. Mark the figure based on the given information.

Marked on diagram.

- d. What other conclusion(s) can be drawn from the information given?

$$\overline{KM} \cong \overline{KM}$$

- e. Explain why this conclusion can be drawn.

Answers will vary. By the reflexive property of congruence, $\overline{KM} \cong \overline{KM}$.



Call attention to the term "bisect." Students may generally understand the term but may forget what occurs when one segment bisects another. Then explain that the definition of a segment bisector or angle bisector results in equality of the bisected parts. To show that these parts are congruent, the definition of congruence should be used.



Consider having students share their rationale from question 1b. If available, use manipulatives to demonstrate the concept of bisecting again and connect that concept to the triangle from this question.



Keep in mind that students may choose to use alternative ways of marking than what is referenced in the answer key here.



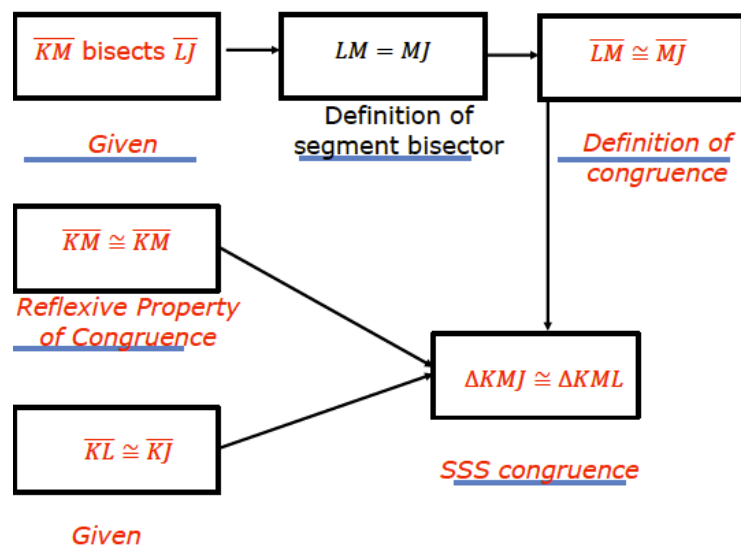
Students may need additional support in completing the proof for this question. Consider walking students through reviewing the conclusions that they have drawn from the question. Assist students with transitioning their logic into a flow proof. Draw attention to what is concluded after given a statement about a bisector. Students may notice that $\triangle KLJ$ is an isosceles triangle, concluding that $\angle L \cong \angle J$. This is an alternate way for students to prove that $\triangle KML \cong \triangle KML$. The flow chart proof in question 1f is laid out for SSS.

5.6.2 Guided Instruction: Proving Triangles Congruent (continued)

- f. Use the conclusions drawn about $\triangle KIJ$ from the information given, along with the figure, to complete the flow chart proof.

Given: $\overline{KM}$ bisects $\overline{LJ}$
 $\overline{KL} \cong \overline{KJ}$

Prove: $\triangle KIJ \cong \triangle KML$



When reviewing the flowchart proof, repeat the rationale on why each of the steps is being identified in the process. Auditory learners can benefit from hearing each step along with the rationale for why those reasons are provided.



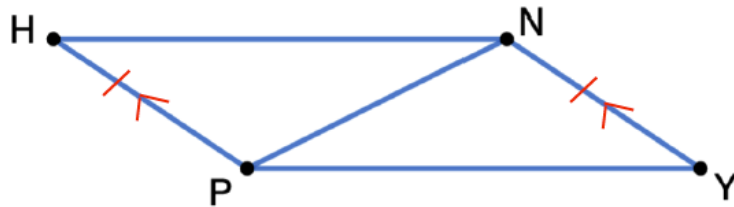
Does the order in which the boxes are completed matter? Sample answer: No. The only time order matters is if the box or boxes has an individual conclusion box following, as the first box in the first column does. The second and third boxes in the first column do not lead to an individual conclusion box — only the final conclusion to prove. Therefore, the second and third boxes in the first column could be swapped.



How could we prove the triangles are congruent using SAS?

5.6.2 Guided Instruction: Proving Triangles Congruent (continued)

2. Polygon $HPYN$ is shown.



- a. Mark the diagram to show the following given information.

Given: $\overline{HP} \cong \overline{YN}$ and $\overline{HP} \parallel \overline{YN}$
Marked on diagram.

- b. Discuss how you determined which markings to make with your partner.
Answers will vary. We selected markings that we could agree on.
- c. Which congruence theorem or postulate should be used to prove $\triangle PHN \cong \triangle NYP$?
Side-Angle-Side
- d. Complete the two-column proof.

Statement	Reason
1. $\overline{HP} \cong \overline{YN}$	1. Given
2. $\overline{HP} \parallel \overline{YN}$	2. <i>Given</i>
3. $\angle HPN \cong \angle YNP$	3. <i>Alternate Interior Angles Postulate</i>
4. $\overline{PN} \cong \overline{PN}$	4. <i>Reflexive Property of Congruence</i>
5. $\triangle PHN \cong \triangle NYP$	5. <i>SAS Congruence Postulate</i>



When reviewing the answers to questions 2a and 2b, consider having students come to the board to make a notation and explain their rationale. Continue this process until all markings have been addressed.



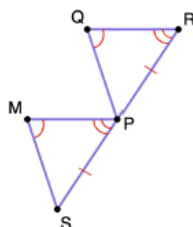
For question 2c, are there any additional congruence theorems or postulates that could be used in this example? Why or why not?



For question 2d, consider using a silent board strategy in which students silently go to the board and fill in one missing statement or reason that they know. This strategy provides an entry point for the students who only know some of the responses, but not all.

5.6.2 Guided Instruction: Proving Triangles Congruent (continued)

3. Triangles MSP and QPR are shown.



Given: $\overline{QR} \parallel \overline{MP}$
 $\angle SMP \cong \angle PQR$
 P is the midpoint of $\overline{SR}$

Prove: $\triangle SPM \cong \triangle PRQ$

- Mark the diagram to show the given information.
Marked on diagram.
- Complete the paragraph proof.

It is given that P is the midpoint of $\overline{SR}$, so $\underline{\overline{SP} \cong \overline{PR}}$

by *definition of midpoint*. It is given that $\overline{QR} \parallel \overline{MP}$,

so $\underline{\angle QRP \cong \angle MPS}$ by *Corresponding Angles Theorem*.

It is also given that $\underline{\angle SMP \cong \angle PQR}$. Therefore,

$\triangle SPM \cong \triangle PRQ$ by *AAS Congruence Postulate*.



If students need additional support, consider creating a word bank for students to use in order to complete the paragraph proof. If you employ this strategy, create additional words that would not be correct for this situation.

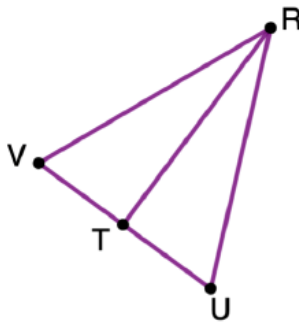
5.6.3 Guided Practice: Proving Triangles Congruent

Component Overview: Students will engage in guided practice with proving triangles are congruent. For this question, review the triangle and given information and allow students to try this on their own. Bring the class back together to review the answers.



5.6.3 Guided Practice: Proving Triangles Congruent

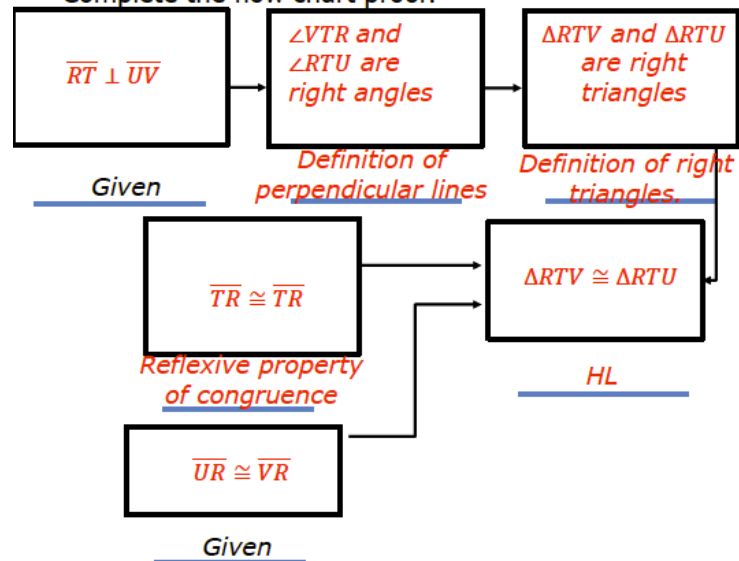
1. Triangle VRU is shown.



Given: $\overline{UR} \cong \overline{VR}$
 $\overline{RT} \perp \overline{UV}$

Prove: $\triangle RTV \cong \triangle RTU$

Complete the flow chart proof.



Think-Pair-Share

For this activity, students can think through their own steps and reasoning and then pair up with a partner to share their answers together to come to a consensus.



For students who finish early, consider having students take the information from a flow chart proof and write it into a paragraph proof for additional practice.

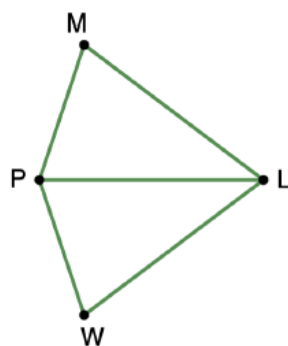
5.6.4 Your Turn!

Component Overview: Students will practice proving two triangles are congruent on their own. For this activity, release students to complete it on their own.



5.6.4 Your Turn!

1. Polygon $MLWP$ is shown.



Given: $\overline{LP}$ bisects $\angle MLW$
 $\angle LPM \cong \angle LPW$

Prove: $\triangle WPL \cong \triangle MPL$

Complete the paragraph proof.

It is given that $\overline{LP}$ bisects $\angle MLW$, so $m\angle MLP = m\angle WLP$ by

definition of angle bisector therefore

$\angle MLP \cong \angle WLP$ by definition of congruence. Also, it

is given that $\angle LPM \cong \angle LPW$.

$\overline{PL} \cong \overline{PL}$ by reflexive property of congruence.

Therefore, $\triangle WPL \cong \triangle MPL$ by ASA congruence.



When reviewing the answer to this question, consider having students do a relay-style review. The relay could be done in the following order:

- Student 1 – Mark the image
- Student 2 – Complete first fill-in-the blank of the paragraph proof
- Student 3 – Complete second fill-in-the blank of the paragraph proof, reading from the beginning through their answer
- Student 4 – Complete third fill-in-the-blank of the paragraph proof, reading from the beginning through their answer
- Student 5 – Complete fourth fill-in-the-blank of the paragraph proof, reading from the beginning through their answer
- Student 6 – Complete last fill-in-the-blank of the paragraph proof, reading from the beginning through their answer
- Student 7 – Read entire proof and defend/illustrate with the shape markings

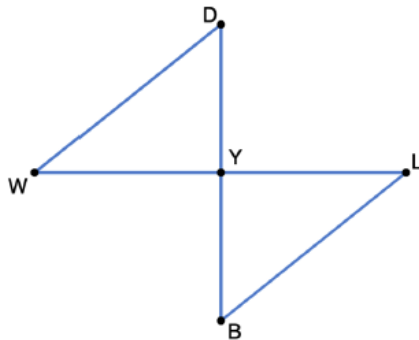
5.6.5 Wrap-Up: Error Analysis

Component Overview: Students will complete an error analysis of a flow chart proof of proving two triangles are congruent. This activity should be done independently to better assess each student's level of understanding of the content.



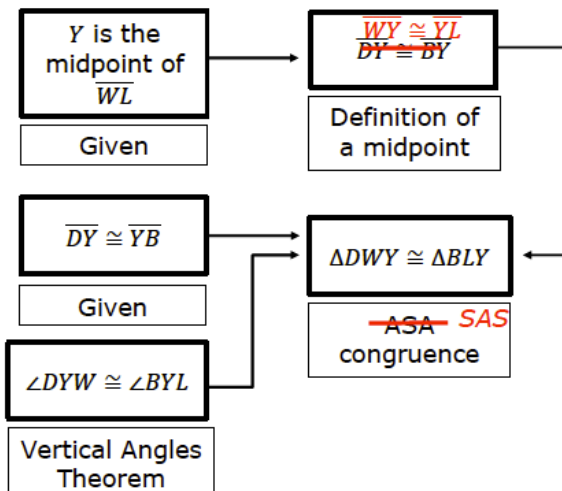
5.6.5 Wrap-Up: Error Analysis

- The flow chart proof has two errors. Find the errors and correct them.



Given: Y is the midpoint of $\overline{WL}$
 $\overline{DY} \cong \overline{YB}$

Prove: $\triangle DWY \cong \triangle BLY$



This question assesses student understanding on proving two triangles are congruent. There are two potential scenarios that may arise from this data:

- Students do not have mastery of triangle congruency
- Students do not have mastery of completing proofs

Use student performance in this activity along with their performance in prior lessons to help determine which area students fall in and where additional support may be needed.